

Analysis of Scientific Data

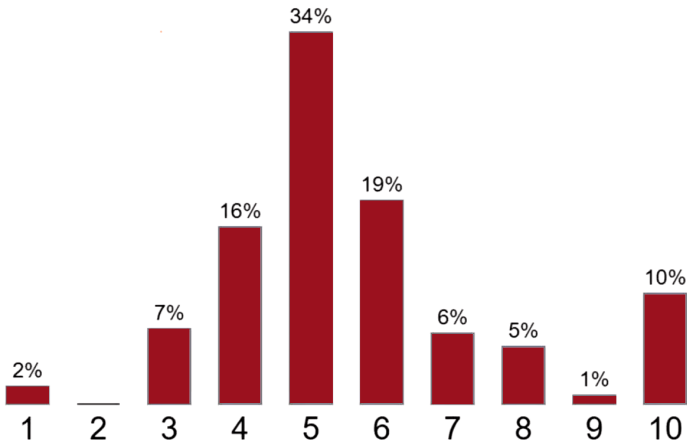


Lecture ~~12~~ 13

Revision

Suspicious?

How many heads in a row would make you suspicious?



Recall that the p -value is a transformation of the data:

$$\text{Data} \longrightarrow \bar{X} \longrightarrow \underline{T} \longrightarrow \underline{P}$$

More generally we have seen

$$\text{Data} \longrightarrow \underbrace{Z}, \underbrace{T}, \underbrace{F}, \underbrace{\chi^2}, \underbrace{\text{Binom}} \longrightarrow \underbrace{P}$$

Deciding how to get from the data to the conclusion becomes important.

Data Types and Analysis

What data types can we analyse with the methods we have learnt in this course?

 <u>Response Variable</u>	<u>Explanatory Variable(s)</u>	<u>Analysis</u>
 Quantitative	—	One-sample t-test Sign test Signed-rank test
 Quantitative	Categorical (2 levels)	Two-sample t-test Rank-sum test
 Quantitative	Categorical (2+ levels)	One-way ANOVA (F-test) Kruskal-Wallis test
 Quantitative	Categorical x Categorical	Two-way ANOVA (F-test)
 Quantitative	Categorical x Quantitative	Multiple regression (t-test)



Quantitative

Quantitative

Simple linear regression (t-test)

Pearson correlation test

Spearman rank test



Quantitative

Quantitative x Quantitative

Multiple regression



Categorical (2 levels)
(2+ levels)

—

Proportion test (Z test)

χ^2 -test



Categorical

Categorical

Two-sample proportion test

χ^2 -test (association)



Categorical (2 levels)

Quantitative

Logistic regression (Z test)

Practice Exam Question 1(e)

Which of the following tests is most appropriate for the data?

1. t -test
2. sign test
3. signed-rank test

Practice Exam Question 2(b)

How many participants were in the study?

1. 63
2. 64
3. 65
4. 66
5. 67
6. 68

Practice Exam Question 3(d)

Should the researchers recommend that young adults reduce their social media usage to improve their sleep quality?

1. Yes
2. No
3. Unsure